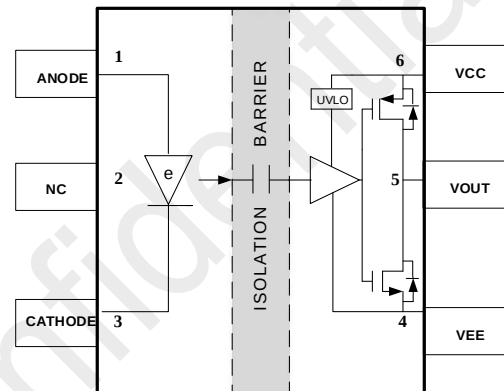


CMT8603X Single-Channel Isolated Gate Driver

Features

- Safety-related certifications
 - DIN VDE V 0884-11: 2017-01
 - UL recognition: up to 5700 Vrms for 1 minute in SOW6;
 - CSA component notice 5A qualification
 - CQC approval per GB4943.1-2022
- Isolated single-channel driver
- Driver side supply voltage: up to 36V with UVLO
- Pin compatible, drop in upgrade for opto isolated gate drivers
- 4A peak source and 6A sink output current
- High CMTI: ± 150 Kv/us
- 75ns typical propagation delay
- 30ns maximum pulse width distortion
- Operation temperature: - 40°C to 125°C
- Lead-free component, suitable for lead-free soldering profile: 260°C, MSL3
- RoHS & REACH Qualified: SOW6

Functional Block Diagram



CMT8603X SOW6 Block Diagram

Applications

- DC to AC solar inverters
- Motor drivers
- UPS and battery charges
- Isolated DC/DC and AC/DC power supplies

Description

The CMT8603X is a single-channel isolated gate driver which is pin-compatible for popular opto-coupled gate driver. It can source 4A and 6A sink peak current. System robustness is supported by 150kV/us minimum common-mode transient immunity (CMTI).

The driver operates with a maximum supply voltage of 32V. While the input circuit imitates the characters of LEDs, it has performance advantages compared to standard opto-isolated gate drivers including better reliability and aging performance, higher working temperature, shorter propagation delay and smaller pulse width distortion.

As a result, the CMT8603X is suitable to replace opto-isolated driver in high reliability, power density and efficiency switching power system.

The CMT8603X is available in SOW6 package.

Device Information

Part No.	Package	Body Size (mm x mm)
CMT8603X-WF	SOW6	7.5mm * 4.68mm
Refer to section 9 for ordering information.		

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1 Absolute Maximum Ratings

Table 1-1. Absolute Maximum Ratings

Parameters	Symbol	Min.	Max.	Unit
Average Input Current	I_{F_AVG}		25	mA
Peak Transient Input Current	I_{F_PEAK}		0.2	A
Reverse Input Voltage	V_{R_MAX}		6.5	V
Driver Side Supply Voltage	$V_{CC} - V_{EE}$	- 0.3	36	V
Output Signal Voltage	V_{OUT}	$V_{EE} - 0.3$	$V_{CC} + 0.3$	V
Operating Junction Temperature	T_J	- 40	150	°C
Storage Temperature	T_{stg}	- 65	150	°C

2 Pin Description

The pin list is shown as below.

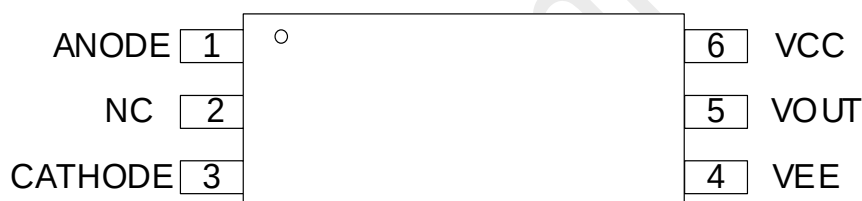


Figure 2-1. CMT8603X SOW6 Top View

Table 2-1. Pin Description

Symbol	Pin No.	Description
ANODE	1	Anode of LED emulator
CATHODE	3	Cathode of LED emulator
VCC	6	Positive output voltage
VOUT	5	Gate-drive output
VEE	4	Negative output voltage
NC	2	No Connection

3 Typical Application

3.1 Typical Application Schematic

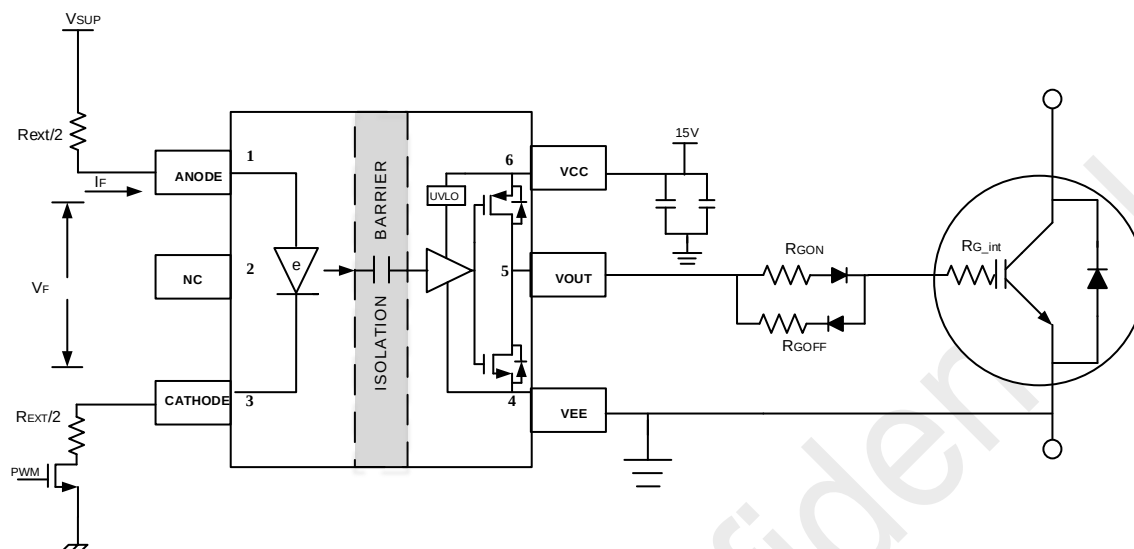


Figure 3-1. Typical Application Circuit with NMOS Driving Input Stage

Bypassing capacitors connecting between V_{CC} and V_{EE} are needed to achieve reliable performance. To filter noise, 0.1 μ F/50V ceramic capacitor is recommended to place as close as possible to CMT8603X. To support high peak currents when turning on external power transistor, additional 10 μ F/50V ceramic capacitor is recommended. If the V_{CC} power supply is located long distance from the IC, bigger capacitance is needed.

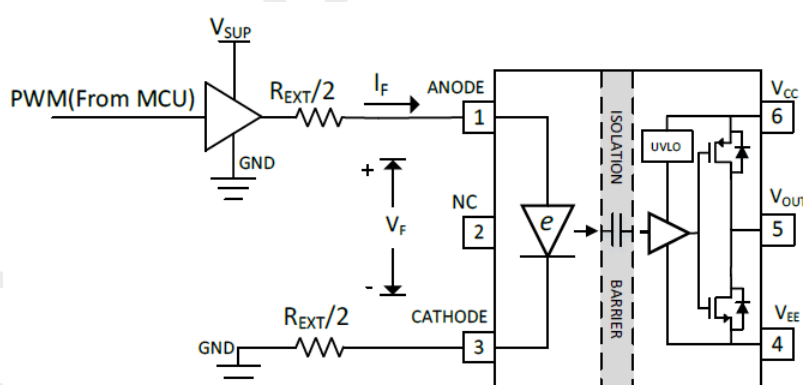


Figure 3-2. Typical Application Circuit with One Buffer Driving Input Stage

CMT8603X requires 7mA to 16mA bias current that flows into the e-diode for normal operation. The PWM from MCU is not suitable to provide such current directly and external circuit is needed. In Figure 3-1, one NMOS is used with split input resistors. Another input drive method is using one buffer, as shown in Figure 3-2.

3.2 Interlock Protection

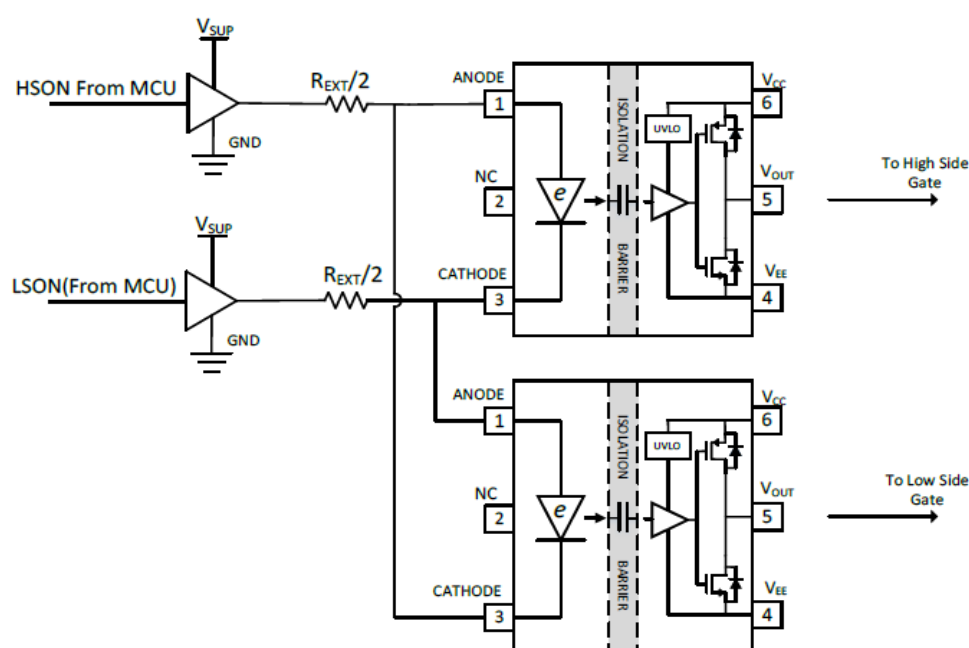


Figure 3-3. Interlock Protection using CMT8603X

For applications to drive power transistors in half bridge configuration, two CMT8603X can be used. Interlock protection is possible as shown in Figure 3-3. If the controller has some mistake, leading to negative dead time, the output PWM of CMT8603X is adjusted to avoid power transistor shoot through. The input side reverse breakdown voltage of CMT8603X is greater than 6.5V, which supports interlock protection of 3.3V or 5V PWM signal source.

3.3 Selecting Input Resistor

The recommended forward current range for CMT8603X is 7mA to 16mA. The value of input resistor, buffer supply voltage and buffer internal resistance influence the forward current, as shown in Equation (1). In Figure 3-1, R_{Buffer} is the on-resistance of the external NMOS. In Figure 3-2, R_{Buffer} is the buffer output impedance in output “High” state. In Figure 3-3, R_{Buffer} is the summary of buffer output impedance in “High” and “Low” state.

$$R_{\text{EXT}} = (V_{\text{SUP}} - V_F) / I_F - R_{\text{Buffer}} \quad (1)$$

The parameter variation needs to be taken into consideration when selecting input resistor.

Table 3-1. External Parameters Range when Calculating Input Resistor

Parameters	Min	Typ	Max	Unit
CMT8603X forward current I_F	7	10	16	mA
CMT8603X forward voltage V_F	1.8	2.1	2.4	V
Buffer supply voltage V_{SUP}	5*95%	5	5*105%	V
Buffer internal resistance R_{buffer}	13	18	22	Ω
External resistance R_{buffer}	203	272	314	Ω

3.4 PCB Layout Guidelines

Careful PCB layout is essential for optimal performance. Some key guidelines are:

- The bypass capacitors should be placed close to CMT8603X, between VCC to VEE.
- There is high switching current that charges and discharges the gate of external power transistor, leading to EMI and ring

issues. The parasitic inductance of this loop should be minimized, by decreasing loop area and place CMT8603X close to power transistor.

- Place large amount of copper connecting to VEE pin and VCC pin for thermal dissipation, with priority on VEE pin. If the system has multiple VEE or VCC layers, use multiple vias of adequate size for connection.
- To ensure isolation performance between primary and secondary side, the space under the chip should keep free from planes, traces, pads or via.

4 Parameter Measurement Circuit Setup

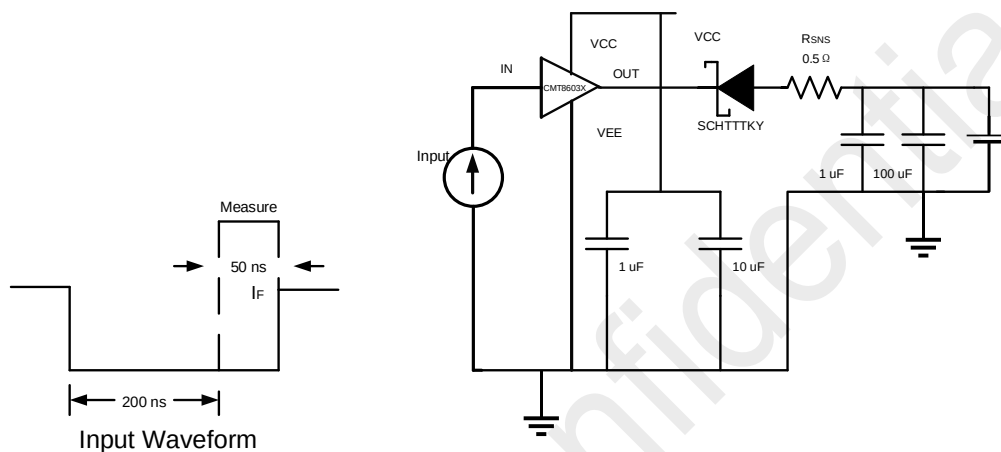


Figure 4-1. I_{OL} Sink Current Test Circuit

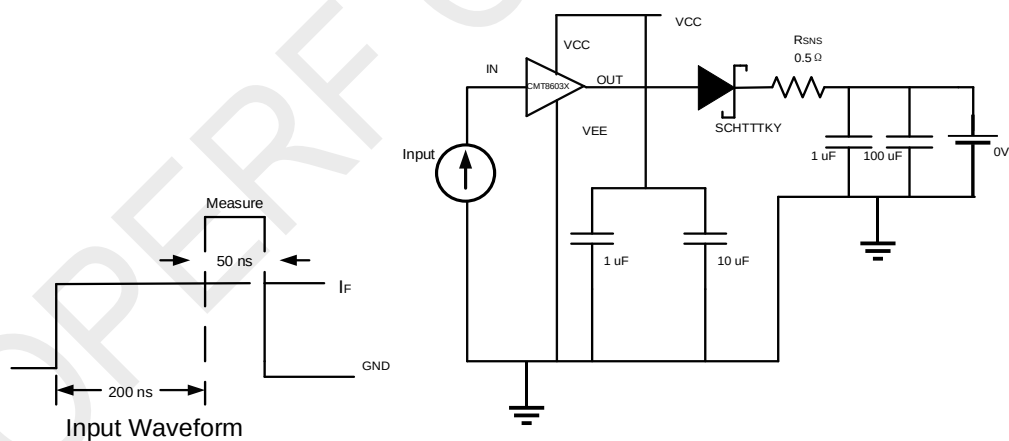


Figure 4-2. I_{OH} Source Current Test Circuit

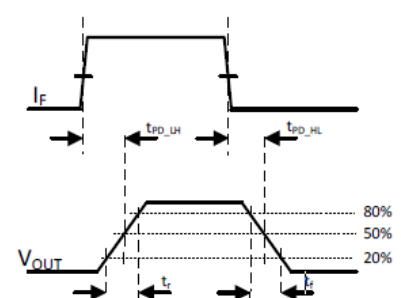
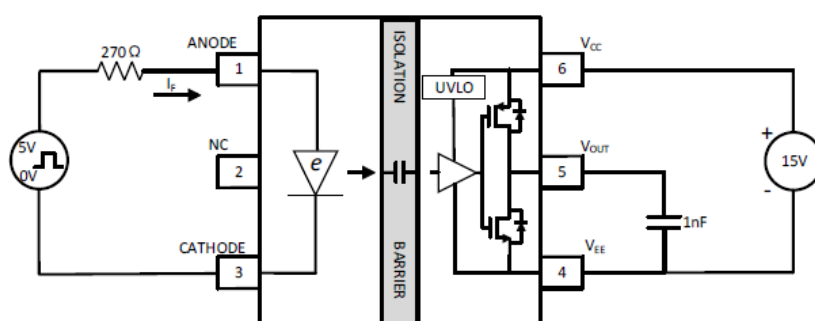


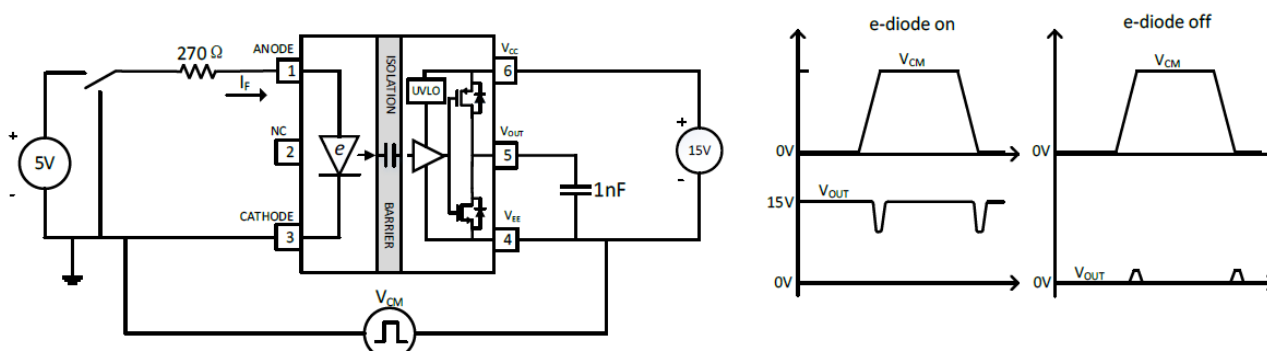
Figure 4-3. I_F to V_{OUT} Propagation Delay, Rise Time and Fall Time

Figure 4-4. Common Mode Transient Immunity Test Circuit

5 Specifications

5.1 DC Electrical Characteristics

All min and max specifications are at $T_J = -40^{\circ}\text{C}$ to 150°C , $V_{CC} = 14\text{V} - 32\text{V}$, $V_{EE} = \text{GND}$, $I_{F(\text{ON})} = 7\text{mA} - 16\text{mA}$, $V_{F(\text{OFF})} = -5.5\text{V} - 0.9\text{V}$. Unless otherwise noted, typical values are at $V_{CC} = 15\text{V}$, $V_{EE} = \text{GND}$, $T_A = 25^{\circ}\text{C}$.

Table 5-1. Electrical Characteristics

Parameters	Symbol	Condition	Min.	Typ.	Max.	Unit
Driver Side Supply						
High level supply current	I _{CCH}	I _F = 10mA, I _{OUT} = 0mA		1.7	3	mA
Low level supply current	I _{CCL}	V _F = 0V, I _{OUT} = 0mA		1.6	3	mA
Driver Side Supply UVLO Threshold (CMT8603X, 13V UVLO Level)						
VCC UVLO Rising Threshold	V _{CC_ON}	I _F = 10mA	12.6	13.2	13.7	V
VCC UVLO Falling Threshold	V _{CC_OFF}		11.7	12.3	12.7	
VCC UVLO Hysteresis	V _{CC_HYS}			0.9		
Driver Side Supply UVLO Threshold ((CMT8603X, 9V UVLO Level)						
VCC2 UVLO Rising Threshold	V _{CC2_ON}	I _F = 10mA	8.7	9.2	9.7	V
VCC2 UVLO Falling Threshold	V _{CC2_OFF}		8	8.5	9	
VCC2 UVLO Hysteresis	V _{CC2_HYS}			0.7		
Input Pin Characteristic						
Input Forward Threshold Current Low to High	I _{FLH}	VO _{UT} > 5V, C _g = 1nF	1.5	2.7	4	mA
Threshold Input Voltage High to Low	V _{FHL}	VO _{UT} < 5V, C _g = 1nF	0.9			V
Input Forward Voltage	V _F	I _F = 10mA	1.8	2.1	2.4	V
Temp Coefficient of Input Forward Voltage	ΔV _F /ΔT	I _F = 10mA		0.34		mV/°C
Input Reverse Breakdown Voltage	V _R	I _R = 10uA	6.5			V
Input Capacitance	C _{IN}	f = 1MHz		17		pF
Output Pin Characteristic						

High Level Output Voltage	V_{OH}	$I_{OUT} = -50\text{mA}, I_F = 10\text{mA}$	$V_{CC} - 0.3$	$V_{CC} - 0.15$		V
		$I_{OUT} = 0\text{mA}, I_F = 10\text{mA}$		V_{CC}		
Low Level Output Voltage	V_{OL}	$I_{OUT} = 50\text{mA}, V_F = 0\text{V}$		30	65	mV
High Level Peak Output Current	I_{OH}	$V_{CC} = 15\text{V}$, pulse width < 10us		4		A
Low Level Peak Output Current	I_{OL}	$V_{CC} = 15\text{V}$, pulse width < 10us		6		A

5.2 Switching Electrical Characteristics

All min and max specifications are at $T_J = -40^\circ\text{C}$ to 150°C , $V_{CC} = 14\text{V} - 32\text{V}$, $V_{EE} = \text{GND}$, $I_{F(\text{ON})} = 7\text{mA} - 16\text{mA}$, $V_{F(\text{off})} = -5.5\text{V} - 0.9\text{V}$. Unless otherwise noted, typical values are at $V_{CC} = 15\text{V}$, $V_{EE} = \text{GND}$, $T_A = 25^\circ\text{C}$.

Table 5-2. Switching Electrical Characteristics

Parameters	Symbol	Condition	Min.	Typ.	Max.	Unit
Propagation Delay	t_{PLH}	$C_{LOAD} = 1\text{nF}$, $f = 20\text{kHz}$ (50% Duty Cycle)	50	75	100	ns
	t_{PHL}		50	69	100	ns
Pulse Width Distortion, $ t_{PHL} - t_{PLH} $	t_{PWD}			6	30	ns
Propagation Delay Difference Between Any Two Parts ($t_{PHL} - t_{PLH}$) ⁽¹⁾	PDD		-35		35	ns
Output Rise Time (20% to 80%)	t_R			6	20	ns
Output Fall Time (80% to 20%)	t_F			6	20	ns
Common Mode Transient Immunity	CMTI		150			kV/us

(1) The difference between t_{PHL} and t_{PLH} between any two parts under the same test condition, ensured by characterization.

5.3 Insulation Specifications

Table 5-3. Insulation Specifications

Parameters	Sym.	Condition	Value	Unit
External clearance	CLR	The shortest terminal-to-terminal distance through air	8.0	mm
External creepage	CRP	The shortest terminal-to-terminal distance across the package surface	8.0	mm
Distance through insulation	DTI	Minimum internal gap	>30	um
Comparative tracking index	CTI	DIN EN 60112 (VDE 0303-11); IEC 60112	> 600	V
Material group	-	IEC 60664-1	I	-

5.4 Insulation Characteristics

Description	Test Condition	Symbol	Value	Unit
Overvoltage Category per	For Rated Mains Voltage $\leq 600V_{RMS}$		I to IV	

IEC60664-1	For Rated Mains Voltage $\leq 1000V_{RMS}$		I to III	
Climatic Category			40/125/21	
Pollution Degree	per DIN VDE 0110		2	
Maximum Working Isolation Voltage	AC voltage	V_{IOWM}	1500	V_{RMS}
	DC voltage		2121	V_{DC}
Maximum Repetitive Peak Isolation Voltage		V_{IORM}	2121	V_{PEAK}
Apparent Charge	Method B, routine test (100% production) and preconditioning (type test); $V_{ini} = 1.2 \times V_{IOTM}$, $V_{pd(m)} = V_{IORM} \times 1.875$, $t_{ini} = t_m = 1s$	q_{pd}	<5	pC
	Method A, after Environmental Tests Subgroup 1, $V_{pd(m)} = V_{IORM} \times 1.6$, $t_{ini} = 60s$, $t_m = 10s$			
	Method A, after Input and Output Safety Test Subgroup 2 and Subgroup 3, $V_{pd(m)} = V_{IORM} \times 1.2$, $t_{ini} = 60s$, $t_m = 10s$			
Maximum Transient Isolation Voltage	$t = 60s$	V_{IOTM}	8000	V_{peak}
Maximum Withstanding Isolation Voltage	$V_{TEST} = V_{ISO}$, $t = 60s$ (qualification); $V_{TEST} = 1.2 \times V_{ISO}$, $t = 1s$ (100% production)	V_{ISO}	5700	V_{RMS}
Maximum Surge Isolation Voltage	Test method per IEC62368-1, 1.2/50us waveform, $V_{TEST} = V_{IOSM} \times 1.6$	V_{IOSM}	8000	V_{peak}
Isolation Resistance	$V_{IO} = 500V$ at $T_A = T_S = 25^\circ C$	R_{IO}	$>10^{12}$	Ω
	$V_{IO} = 500V$ at $T_A = T_S = 150^\circ C$		$>10^9$	
	$V_{IO} = 500V$ at $100^\circ C \leq T_A \leq 125^\circ C$		$>10^{11}$	
Isolation Capacitance	$f = 1MHz$	C_{IO}	1	pF

5.5 Safety-Limiting Values

Description	Test Condition	Symbol	Value	Unit
Maximum Safety Temperature		T_S	150	$^\circ C$
Safety Input, Output, or Total Power	$R_{\theta JA} = 125^\circ C/W$, $T_J = 150^\circ C$, $T_A = 25^\circ C$	P_S	750	mW
Safety Input, Output, or Supply Current	$R_{\theta JA} = 125^\circ C/W$, $T_J = 150^\circ C$, $T_A = 25^\circ C$	I_S	50	mA
	$R_{\theta JA} = 125^\circ C/W$, $V_{CC} = 30V$,		25	

	$T_J = 150^{\circ}\text{C}, T_A = 25^{\circ}\text{C}$			
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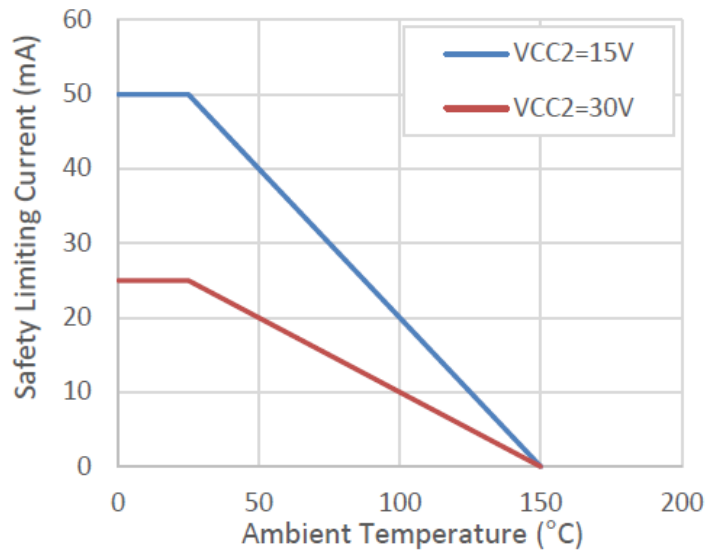


Figure 5-1. Thermal Derating Curve for Limiting Current per DIN VDE V 0884-11

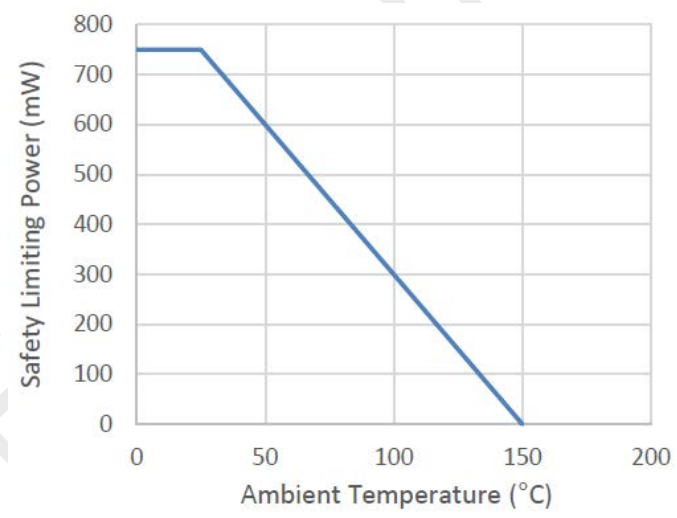


Figure 5-2. Thermal Derating Curve for Limiting Power per DIN VDE V 0884-11

5.6 Typical Performance

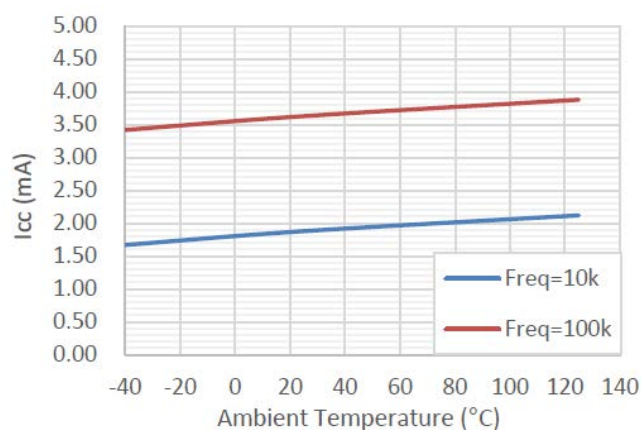


Figure 5-3. Supply Current vs Temperature

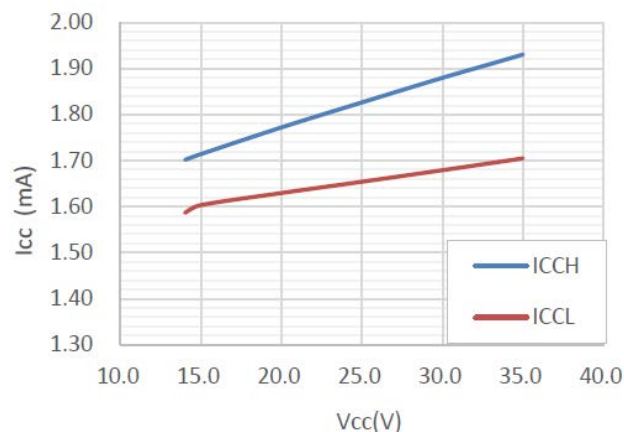


Figure 5-4. Supply Current vs Supply Voltage

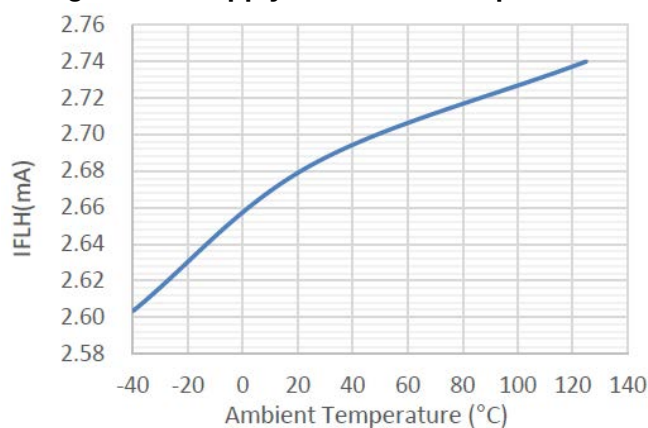


Figure 5-5. Forward Threshold Current vs Temperature

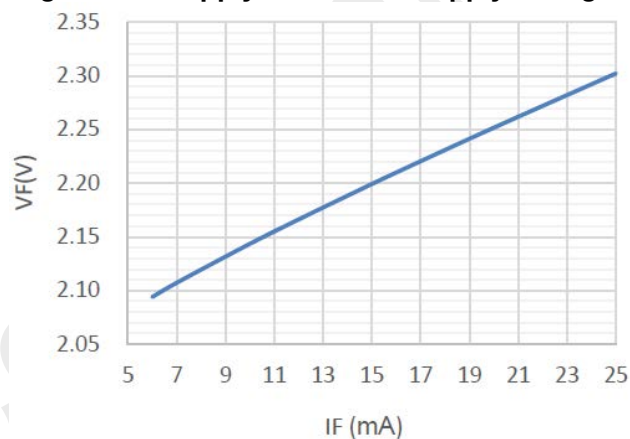


Figure 5-6. Forward Current vs Forward Voltage Drop

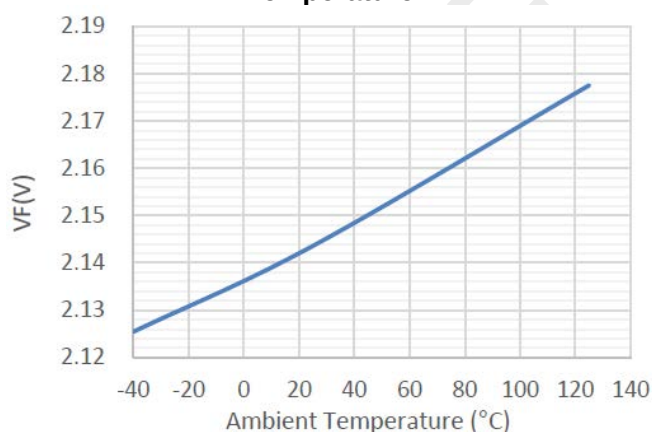


Figure 5-7. Forward Voltage Drop vs Temperature

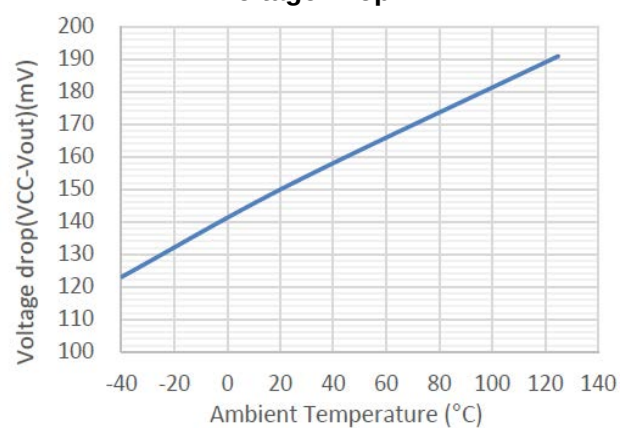


Figure 5-8. V_{OH} (50mA Load) vs Temperature

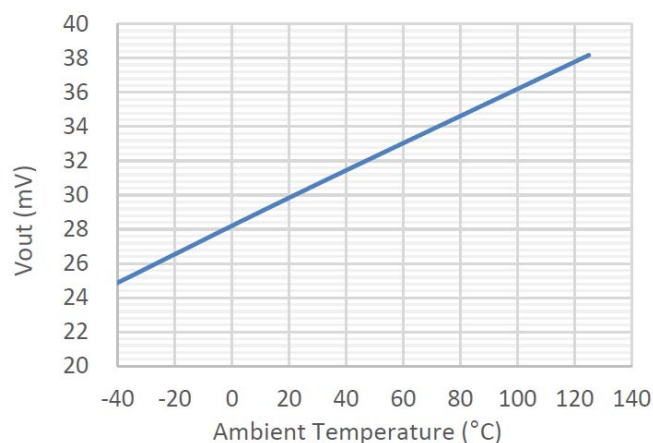
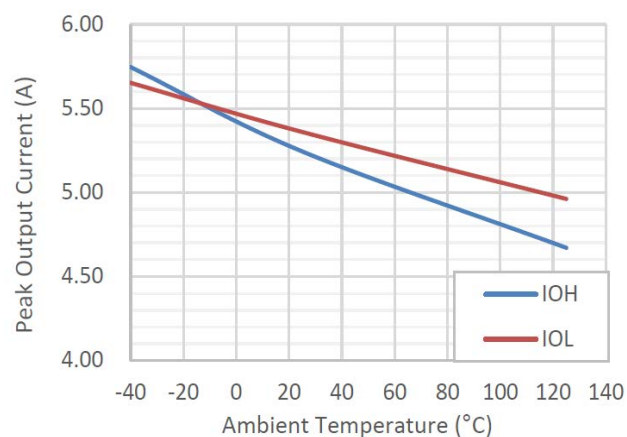
Figure 5-9. V_{OL} vs Temperature

Figure 5-10. Output Drive Currents vs Temperature

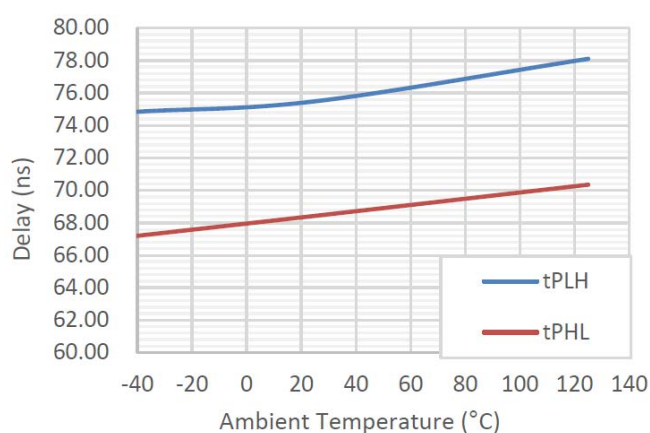


Figure 5-11. Propagation Delay vs Temperature

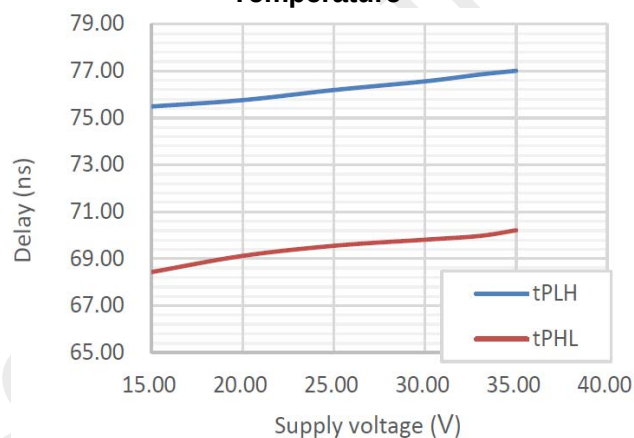
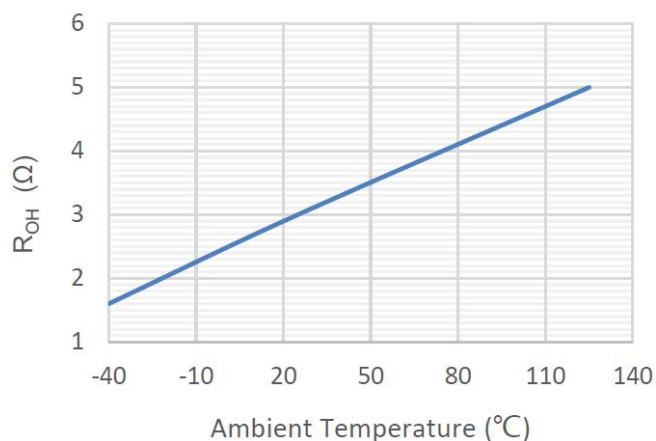
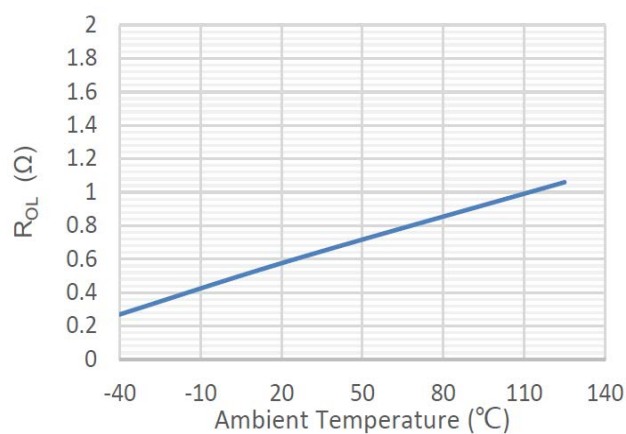


Figure 5-12. Propagation Delay vs Supply Voltage

Figure 5-13. R_{OH} vs TemperatureFigure 5-14. R_{OL} vs Supply Voltage

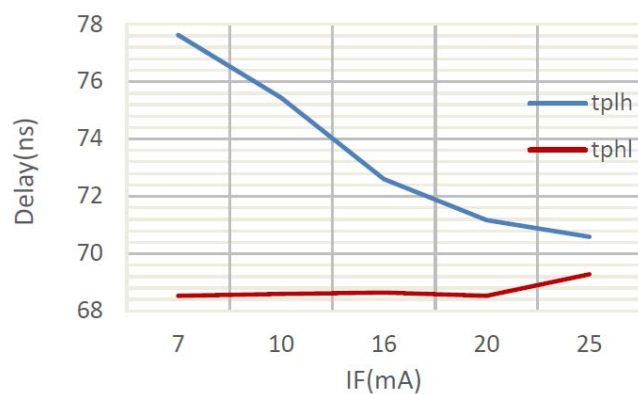


Figure 5-15. Propagation Delay vs Forward Current

6 Safety-related Certifications

6.1 Regulatory Information

UL	CSA	VDE	CQC
UL 1577 Component Recognition Program	Approved under CSA Component Acceptance Notice 5A	Certified according to DIN EN IEC 60747-17(VDE 0884-17)	Certified by GB4943.1-2022
E541580		Pending	CQC23001382478

7 Function Description

7.1 Function Overview

The CMT8603X is a single-channel isolated gate driver which is pin-compatible for popular opto-coupled gate driver. The integrated galvanic isolation between control input logic and driving output stage grants additional safety. The device can source 4A and sink 6A peak current, which can drive IGBTs, power MOSFETs and SiC MOSFETs in many applications such as motor control systems, solar inverters and power supplies.

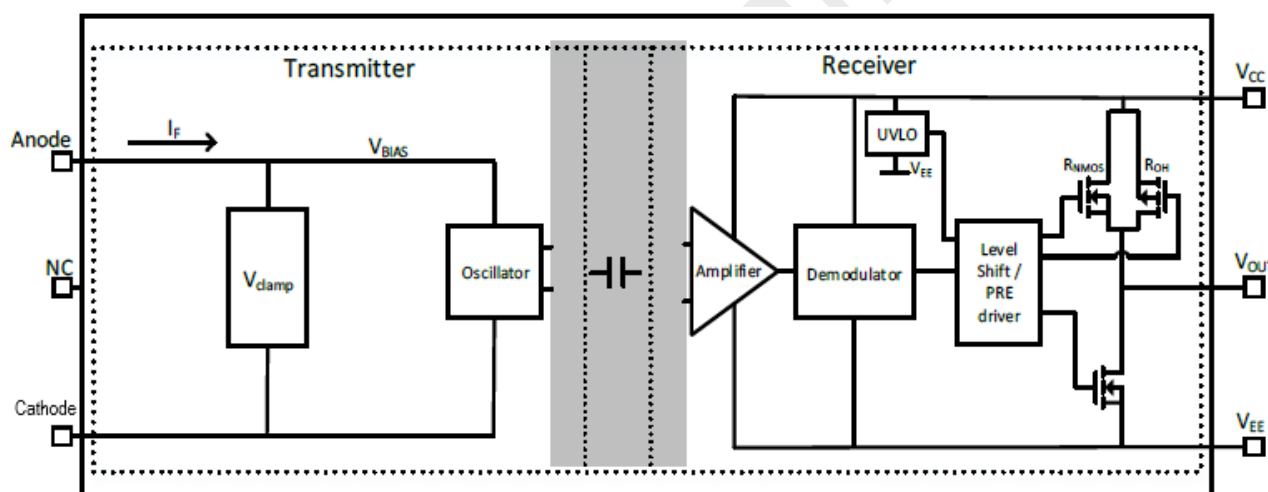


Figure 7-1. CMT8603X Functional Block Diagram

7.2 Truth Tables

Table 7-2. Driver Function Table^[1]

e-diode	V _{CC} status	Outputs
X	Powered down	L
$I_F > I_{FLH}$	Powered up	H
$V_F > V_{FLH}$	Powered up	L

1) H= Logic High; L= Logic Low; X= Irrelevant

7.3 Output Stage

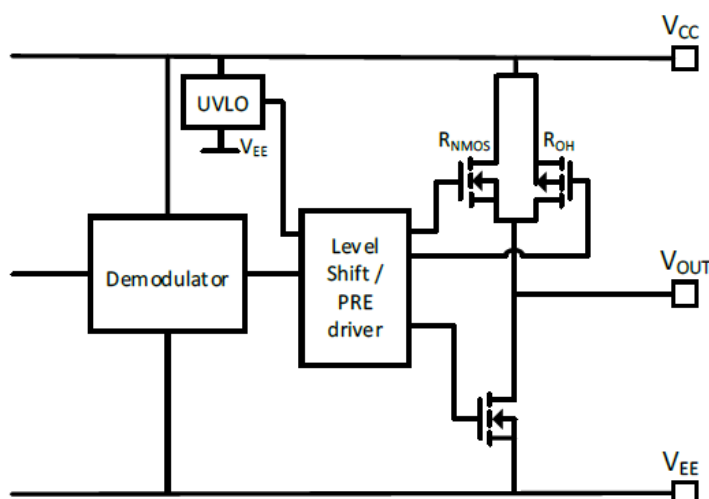


Figure 7-3. CMT8603X Output Stage

Table 7-3. CMT8603X Output Stage On-Resistance

R_{NMOS}	R_{OH}	R_{OL}	Unit
1.4	5	0.6	Ω

The CMT8603X has P-channel and N-channel MOSFET in parallel to pull up the OUT pin when turning on external power transistor. During DC measurement, only the P-channel MOSFET is conducting. The measurement result R_{OH} represents the on-resistance of P-channel MOSFET.

The voltage and current of external power transistor drain to source or collector to emitter change during turn on. At that time, the CMT8603X N-channel MOSFET turns on to pull up OUT more quickly. It results external power transistor faster turn on time, lower turn on power loss, also leads to smaller temperature increase of CMT8603X. The equivalent pull-up resistance of CMT8603X is the parallel combination $R_{OH} \parallel R_{NMOS}$. The result is quite small, indicating the strong driving capability of CMT8603X.

The pull-down structure of CMT8603X is simply composed of an N-channel MOSFET with on-resistance of R_{OL} . The result is quite small, indicating the strong driving capability of CMT8603X.

7.4 VCC and Under Voltage Lock Out (UVLO)

The lower limit of driver side supply voltage (V_{CC}) is determined by the internal UVLO protection feature of the device. V_{CC} voltage should not fall below the UVLO threshold for normal operation, or else the gate-driver outputs can become clamped low.

A local bypass capacitor should be placed between the V_{CC} and V_{EE} pins, with a value of 220nF to 10 μ F for device biasing. An additional 100nF capacitor in parallel with the device biasing capacitor is recommended for high frequency filtering. Both capacitors should be positioned as close to the device as possible. Low-ESR, ceramic surface-mount capacitors are recommended.

7.5 Active Pull-Down

The Active Pull-Down feature ensures a safe IGBT or MOSFET off-state if V_{CC} is not connected to the power supply. When V_{CC} is floating, the driver output is held low and clamping V_{OUT} pin to approximately 1.9V higher than V_{EE} .

7.6 Short Circuit Clamping

During short circuit the gate voltage of IGBT or MOSFET tends to rise because of the feedback via the Miller capacitance. The diode between V_{OUT} and V_{CC} pins inside the driver limits this voltage to a value slightly higher than the supply voltage. A maximum current of 500mA may be fed back to the supply through this path for 10 μ s. If higher currents are expected or tighter clamping is desired external Schottky diodes may be added.

7.7 Thermal Shutdown

Parameter	Symbol	Value	Unit
Junction-to-ambient thermal resistance	$R_{\theta JA}$	125	$^{\circ}\text{C}/\text{W}$
Junction-to-top characterization parameter	Ψ_{JT}	30	$^{\circ}\text{C}/\text{W}$

8 Packaging Information

The packaging information of the CMT8603X is shown in the figures below.

8.1 CMT8603X SOW6 Packaging

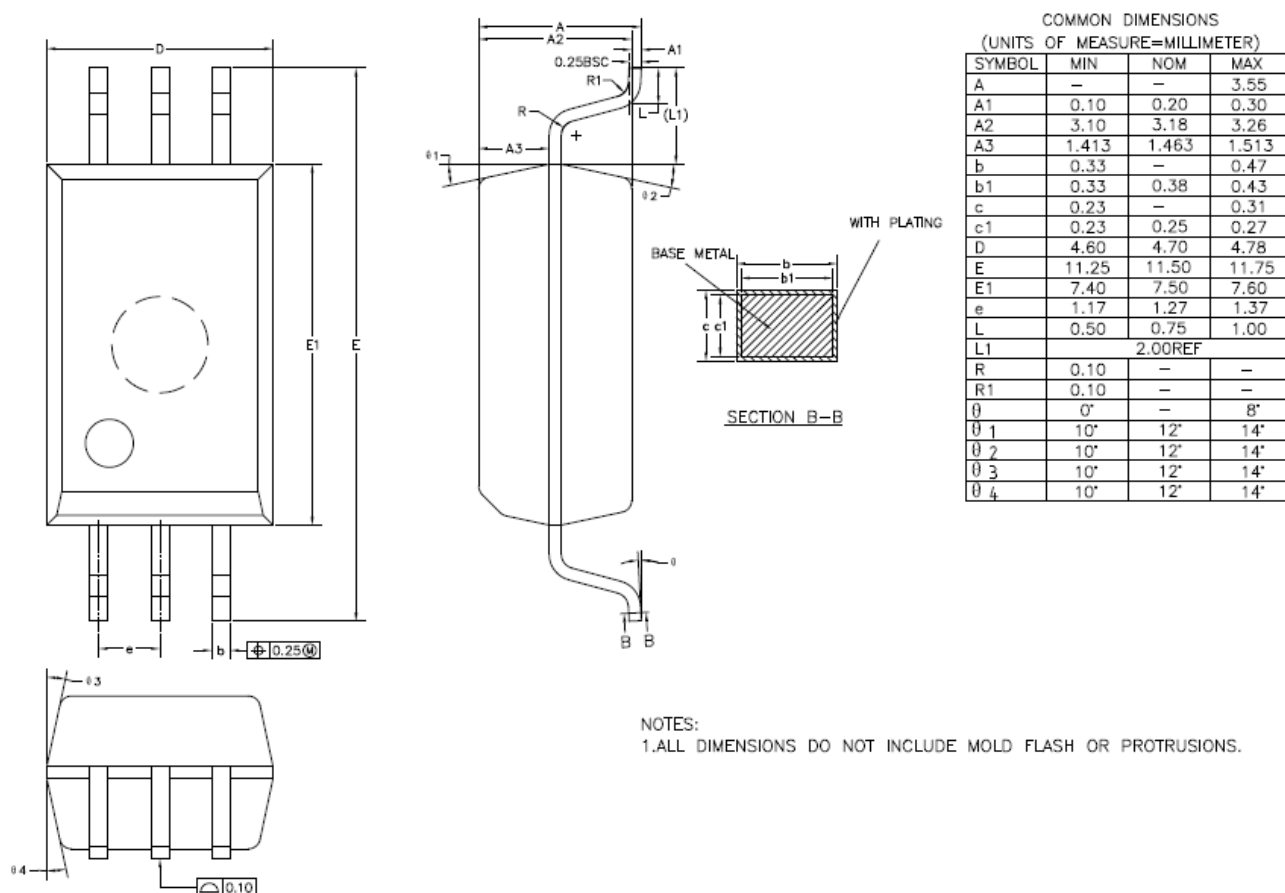


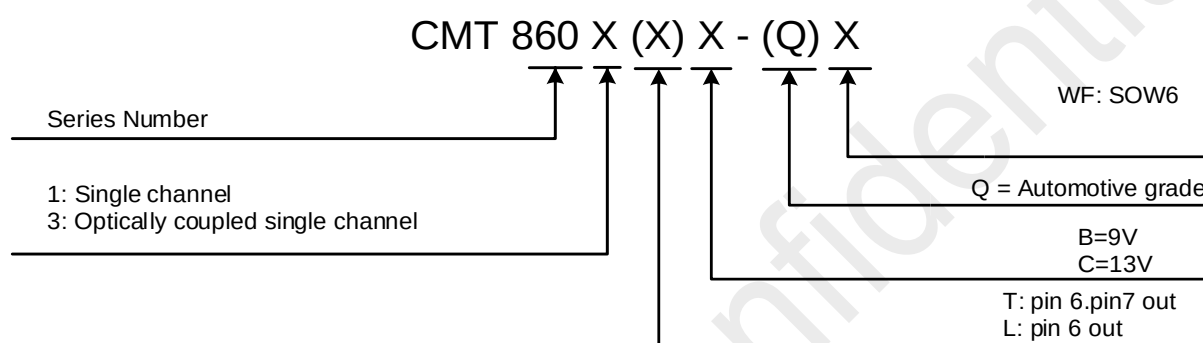
Figure 8-1. SOW6 Packaging

9 Ordering Information

Table 9-1. Part Number List

Part Number	MOQ	Isolation Rating (kV _{rms})	UVLO Level	Vout Connection	Temperature	Package	MSL
CMT8603B-WF	1000	5.7	9	5	-40 to 125°C	SOW6	3
CMT8603C-WF	1000	5.7	13	5	-40 to 125°C	SOW6	3

Part Number Naming Rule:



Please visit www.hoperf.com for more product/product line information.

Please contact sales@hoperf.com or your local sales representative for sales or pricing requirements.

10 Tape and Reel Information

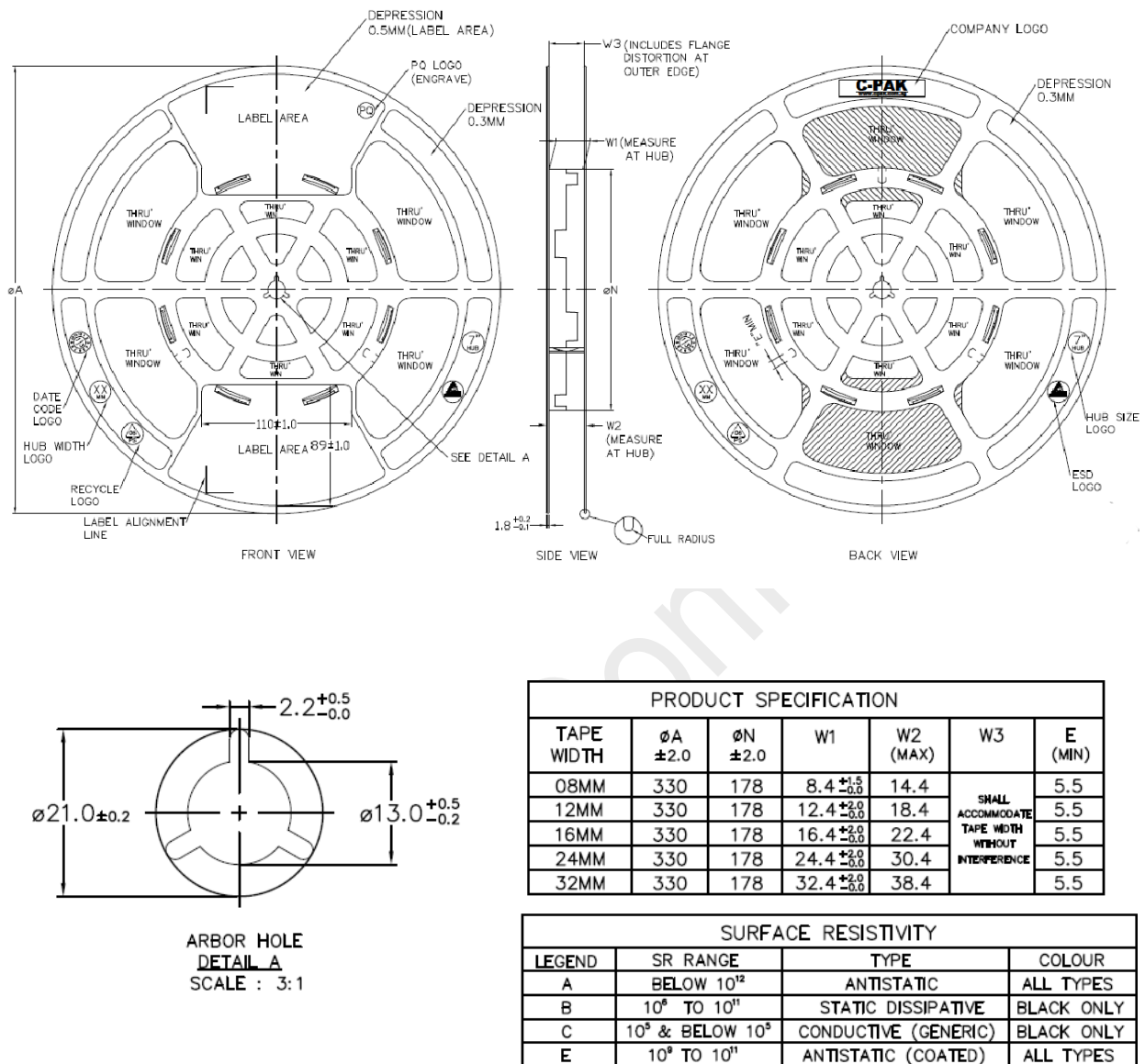


Figure 10-1. CMT8603X Tape and Reel Information

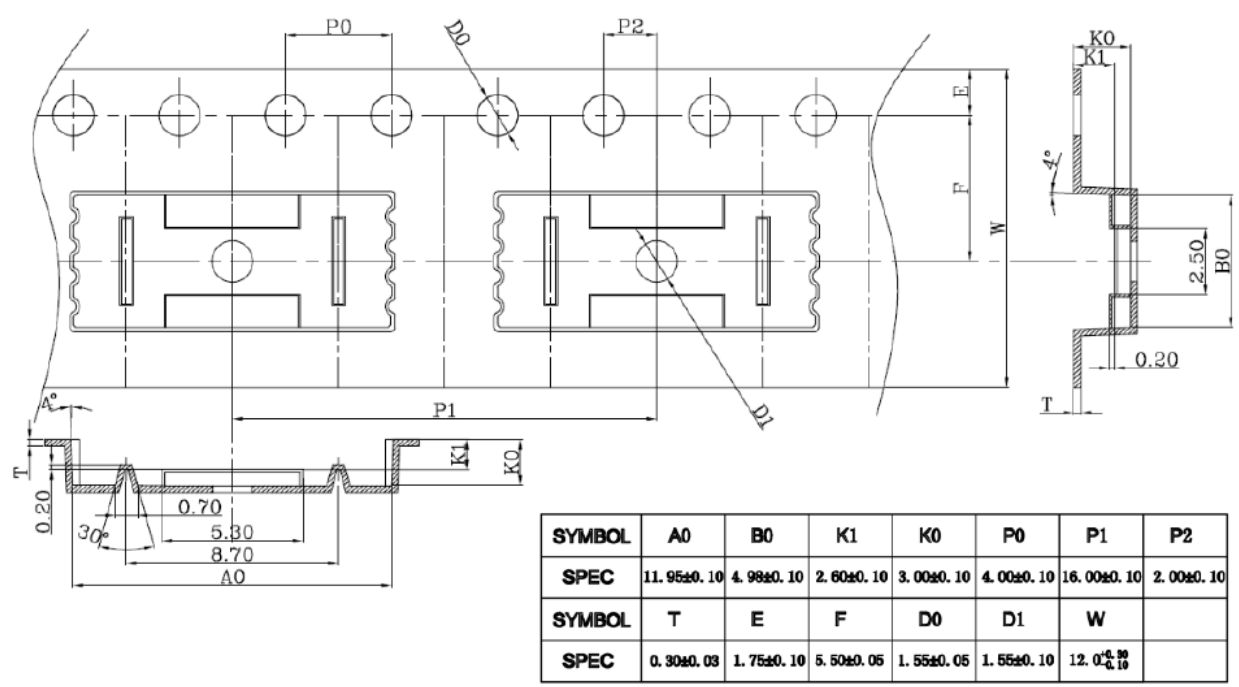


Figure 10-2. Tape Information of SOW6

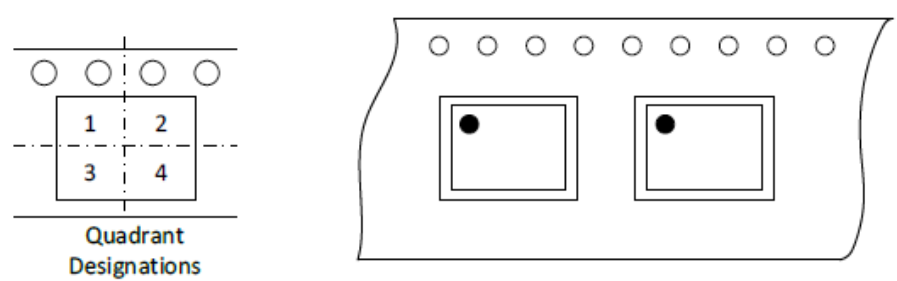


Figure 10-3. Quadrant Designation for Pin1 Orientation in Tape

11 Revise History

Table 11-1. Revise Records

Version No.	Chapter	Description	Date
0.1	All	Initial version	2023/03/29
0.2	All	Update the driver side supply voltage to 36V	2023/6/17
0.3	All	Delete part number CMT8603B-N/CMT8603B-U/CMT8603C-N/CMT8603C-U	2025/6/9
		Update the CQC certificate number	2025/7/3
0.4	All	Revise the package information	2026/1/13

12 Contacts

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